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Algorithms

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Merge Sort: Introduction

Sorting :- arranging the things in increasing
order or decreasing order

B/s $\rightarrow$ $\{10, 5, 6, 26, 1, 3\}$
ALS $\rightarrow$ $[1, 3, 5, 6, 10, 26]$

For sorting there are many algorithms.
merge sort and quick sort are examples of DAC.

Merge + Sort : The actual Meaning

Pahle: sort, then merge

Merge Sort is nothing Just merging two sorted arrays into single sorted array

• Ex:

$a = [2, 5, 9, 10]$
sorted

$b = [4, 6, 27]$
sorted

merge

$c = [$



$a = [2, 5, 9, 10]$
↑ ↑ ↑ ↑
i i i i

$b = [4, 6, 27]$
↑ ↑ ↑
j j j

$c = [1, 2, 5, 6, 9, 10, 27, 40, 46]$
↑ ↑
k k

```

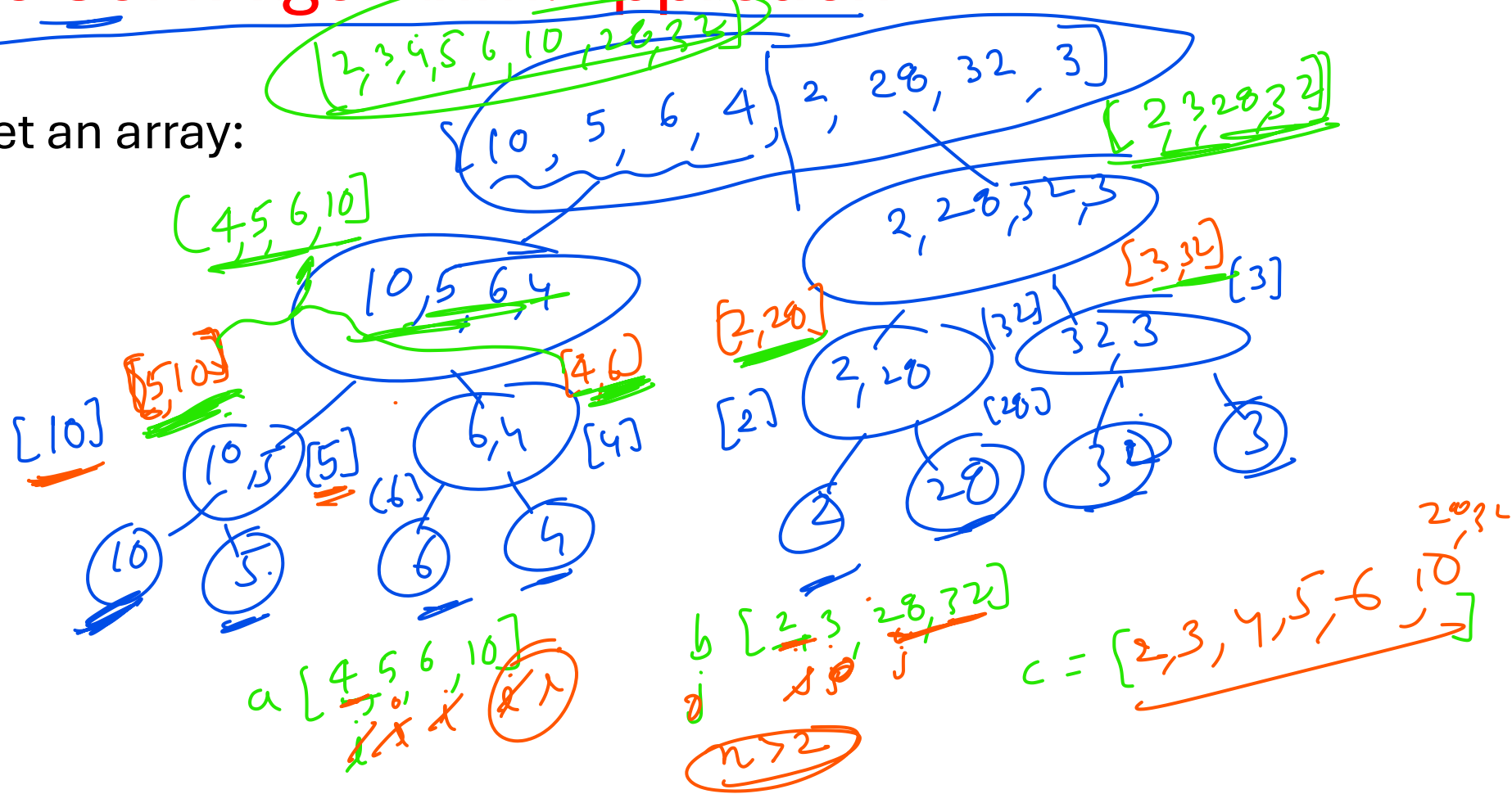
if ( b[j] <= a[i] )
{
    c[k] = b[j]
    k++
    j++
}
if ( a[i] < b[j] )
{
    c[k] = a[i]
    k++
    i++
}

```

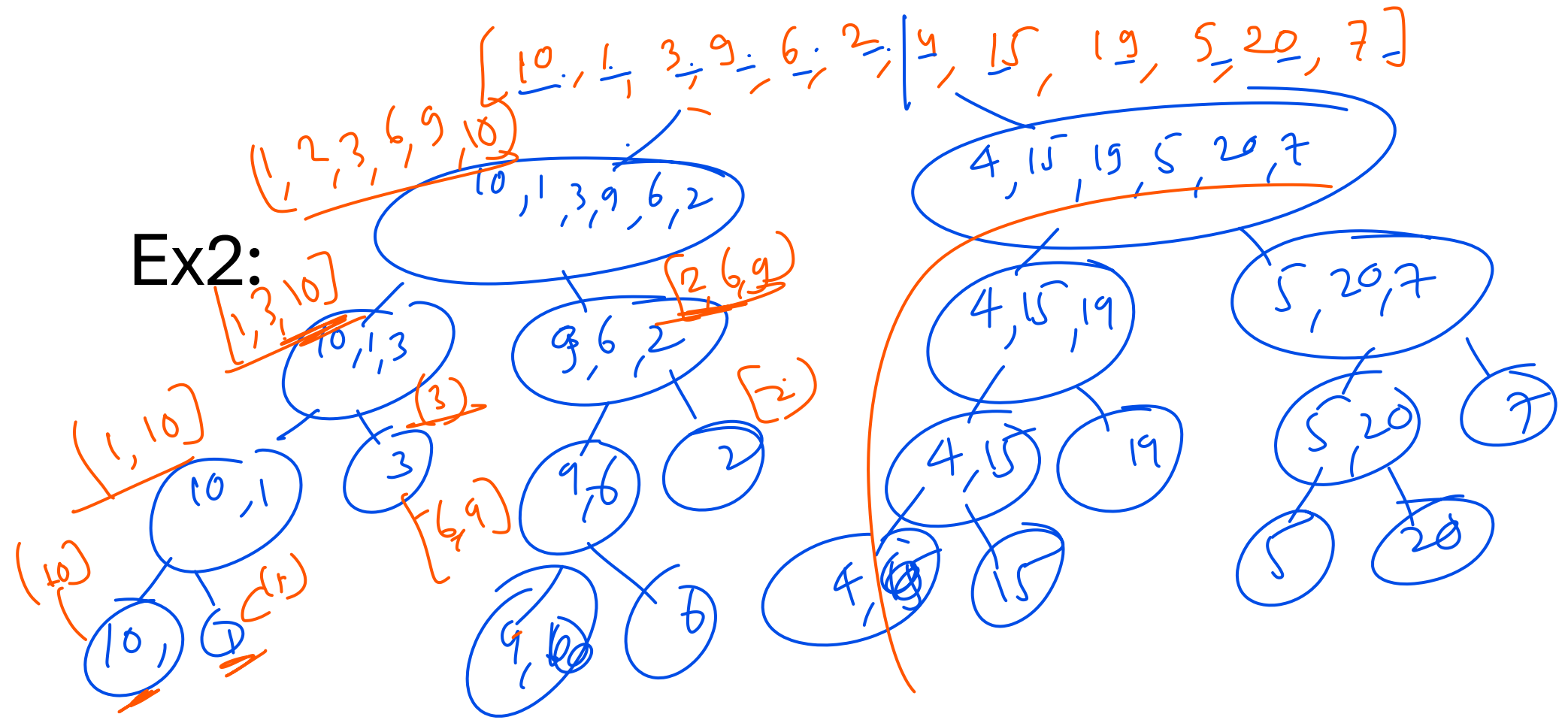
Merge Sort Algorithm: Approach

→ DAC

- Ex1: Let an array:



Ex2:



[10, 1, 3, 4]

mergesort(a, i, j)

{ if (i == j)

return a[i];

mid = (i + j) / 2

mergesort(a, i, mid)

mergesort(a, mid + 1, j)

merge(x, y, i, mid, j)

}



